MM-DeepResearch: Reproduction

A Simple and Effective Multimodal Agentic Search Baseline

Yao et al. (ByteDance, NTU) · ICML 2026 Paper #631 · arXiv:2603.01050

🔧 github.com/erobinson-mlis/repro-mm-deepresearch · 🤗 Logbook

📊 Reported Benchmark Results (Paper Table 1)

SimpleVQA	MMSearch	LiveVQA	FVQA	InfoSeek	BrowseComp-VL
62.6 +18.1	67.8 +30.4	54.3 +11.3	55.4 +15.3	57.6 +15.8	80.4 +11.2

8B Average: 63.0% (+17pp over Qwen3-VL-8B) · **32B Average: 65.3%** (+14.9pp over Qwen3-VL-32B)

C1 Hyper-Search QA

Hypergraph-based QA generation: builds depth-D hypergraphs from web sources, produces intra- and inter-hyperedge QA pairs → **Hyper-Search-3K** dataset.

Supported Methodology well-specified; generation code not open-sourced.

C2 DR-TTS Trajectories

4 tool types trained via RL experts → tree search recomposition → **DR-TTS-10K** SFT trajectories.

- **text_search** (E5 FAISS)
- **image_search_by_text_query** (Jina-CLIP)
- **image_search_by_lens** (image similarity)
- **model_search** (expert LLM)

Supported Tools confirmed; generation code not in repo.

C3 Training Pipeline

CONFIRMED — Full code audit.

- SFT base: `HuanjinYao/MM-DeepResearch-8B-SFT`
- GRPO RL: n=5 rollouts, lr=1e-6, 35 epochs
- Max 4 tool turns, max len 70K
- **Format reward** (0.1): tag structure
- **Accuracy reward** (0.9): EM + LLM judge
- 17 data sources
- 8x H20 GPUs

CONFIRMED

C5 Ablation: Offline Corpus

E5 (text) + Jina-CLIP (image) embeddings with FAISS GPU indexes, matching paper.

Supported Infrastructure confirmed.

C4 Benchmark Results

Paper claims 63.0% (8B) / 65.3% (32B) average accuracy across 6 benchmarks.

Model	Avg	Δ vs base
Qwen3-VL-8B	46.0	—
MM-DeepResearch-8B	**63.0**	**+17.0**
Qwen3-VL-32B	50.4	—
MM-DeepResearch-32B	**65.3**	**+14.9**

Not Verified Needs 2x A100/H100 + search API keys.

Data Resources

Model (8B):
https://huggingface.co/HuanjinYao/MM-DeepResearch-8B
Corpus:
https://huggingface.co/datasets/HuanjinYao/MM-DeepResearch-corpus
GitHub:
https://github.com/HJYao00/MM-DeepResearch

📋 Conclusion

Claim	Status
C1: Hyper-Search	Supported
C2: DR-TTS	Supported
C3: Training	Confirmed
C4: Benchmarks	Paper only
C5: Ablation	Supported

Core training pipeline is fully reproducible. Data generation pipelines not open-sourced. Benchmark verification requires GPU compute + paid API keys.

🏗️ Architecture

- Base: Qwen3-VL-8B/32B-Instruct
- Training: VeRL + Ray + SGLang
- Retrieval: FAISS GPU (E5 + Jina-CLIP)
- Reward: Format + EM + LLM Judge
- Tools: text_search, image_search, lens_search, model_search